SSG AI/ML Trading System

Detailed Technical Documentation

Live Fyers Integration • Machine Learning • Outcome Evaluation • Paper Trading • Dashboard

Primary symbol: NSE:WIPRO-EQ | 1-minute candles | 10-minute prediction horizon

Item	Current implementation
Data source	Fyers WebSocket/API
Primary symbol	NSE:WIPRO-EQ
Candle timeframe	1 minute
Prediction horizon	10 minutes
Target threshold	0.10%
Features	26
Classes	BUY / HOLD / SELL
Production model	HIST_GRADIENT_V2_1
Paper strategy	BUY ONLY
Paper quantity	100 shares
Minimum paper confidence	60%
Real-money trading	OFF

Document Purpose

This document is written to address the review requirements that the system demonstrate live Fyers integration, a genuine machine-learning workflow, measurable evaluation on actual market data, next-day comparison, end-to-end background processing, and sufficient technical understanding to explain the implementation. It is intentionally implementation-focused rather than a theoretical research paper.

1. Assignment Requirements and Response

Reviewer requirement	How the project addresses it	Evidence available
Live Fyers data	Fyers WebSocket receives live ticks and writes them into SQLite.	Live collector output with connected socket, subscription and WIPRO ticks.
Background live processing	Collector, candle builder, predictor, evaluator and paper engine run as separate processes.	Terminal outputs observed during live market session.
Machine learning implementation	26-feature supervised classification model with BUY/HOLD/SELL target and predict_proba confidence.	Production model loads as HistGradientBoostingClassifier; 26 features verified.
Actual performance measurement	Predictions are persisted and later evaluated against future ten-minute price movement.	Evaluator moved V2.1 records from pending to evaluated.
One full day + next day	The database stores timestamped predictions and outcomes across multiple	Current data spans 2026-08-31 and 2026-09-01; exact daily metrics must be extracted before final submission.

	sessions, enabling day-by-day analysis.	
Live evidence	The dashboard is only a view; core processing occurs in backend processes.	Collector/candle/predictor/evaluator/paper terminals.
15-minute video	A detailed recording sequence is defined in Section 17.	Video should be recorded from the live system.
Independent implementation	The implementation was debugged and validated through live-token, process, stale-feed, database and model checks.	Development troubleshooting and runtime validation evidence.

2. Problem Definition

The objective is to estimate the short-term directional class of the next ten-minute price move for WIPRO using live market data. The system is not designed to forecast an exact future price. Instead, it converts the future return into three classes: BUY, SELL, or HOLD.

This framing converts the task into supervised multiclass classification. A prediction is useful only if the class is later compared against the real market outcome. Therefore, the project stores every prediction with its timestamp, class, confidence and entry price so that a future outcome can be evaluated without changing the original prediction.

3. Overall System Architecture

Fyers WebSocket
↓
Live Tick Collector
↓
SQLite ticks table
↓
1-minute Candle Builder
↓
Completed Candles
↓
Feature Engineering (26 features)
↓
HIST_GRADIENT_V2_1
↓
BUY / HOLD / SELL + confidence
↓
ml_predictions table
↓
10-minute Outcome Evaluator
↓
Paper Trading Engine

↓

Streamlit Dashboard

The critical point for the assignment is that the dashboard is not the source of truth. The live system continues to run through backend processes even when the dashboard is closed.

4. Live Fyers Integration

4.1 Authentication

The project uses the Fyers authentication-code flow. A fresh auth_code is generated after browser authentication and exchanged for an access token using auth/exchange_token.py. The script explicitly reads the access_token field returned by Fyers and writes it into the local configuration. The refresh token is not used as the WebSocket access token.

python -m auth.fyers_auth

python -m auth.exchange_token

During validation, an earlier WebSocket subscription attempt returned an invalid-token error. The token was re-generated and exchanged. A subsequent live connection authenticated successfully and received continuous WIPRO ticks. This is practical evidence that the integration was tested against the real Fyers endpoint rather than a mocked feed.

4.2 WebSocket Data

The collector receives live market fields including:

- Last traded price (LTP)
- Last traded quantity (LTQ)
- Bid price and quantity
- Ask price and quantity
- Exchange feed timestamp

The collector uses the exchange timestamp when available and stores the records in the SQLite ticks table.

4.3 Live Feed Health

A watchdog monitors the age of the latest WIPRO tick. During testing, a stale-feed condition was detected after approximately 30 seconds without a new tick. The controlled recovery path closed the stale socket. This behavior is intentionally treated as a feed-health problem, distinct from an authentication problem.

5. Tick Storage and Candle Construction

Each tick is persisted first. The candle builder aggregates ticks into one-minute OHLCV records. This creates a reproducible intermediate dataset that can be inspected independently of the ML layer.

Candle field	Meaning

Open	First observed traded price during the minute
High	Maximum traded price during the minute
Low	Minimum traded price during the minute
Close	Last traded price during the minute
Volume	Aggregated traded quantity used by the candle builder

The live predictor deliberately uses completed candles. If the current minute is still forming, it is excluded from prediction. This reduces the risk of using incomplete information.

5.1 Candle Continuity Protection

A data-quality issue identified during validation was a long gap between candle timestamps. For example, the live dataset showed a gap from 13:13 to 14:35 on 2026-08-31. The V2.1 predictor was therefore enhanced with a continuity check. For a 1-minute strategy, a gap greater than one minute causes the prediction to be skipped until the candle stream is continuous again.

MAX_CANDLE_GAP_MINUTES = 1

This is a safety measure. It is preferable to miss a signal rather than generate a signal from a discontinuous feature sequence.

6. Technical Feature Engineering

The production predictor uses 26 features. These features are computed from the candle history and are grouped into momentum, volatility, candle structure, volume, SMMA relationships, trend and recent-range positioning.

Feature group	**Examples**	**Interpretation**
Returns	return_1, return_3, return_5, return_10, return_15	Measures recent percentage price movement over several windows.
Momentum	momentum_acceleration	Measures change in short-horizon momentum rather than only the current return.
Volatility	volatility_5, volatility_20, volatility_ratio	Captures recent variability and short/long volatility regime.
Candle structure	body_ratio, upper_wick_ratio, lower_wick_ratio, range_pct	Describes candle body, rejection wicks and range size.
Volume	relative_volume, volume_ratio_5, volume_ratio_20_change	Compares current volume with rolling averages.
SMMA position	price_smma20_ratio, price_smma120_ratio, smma_spread	Describes price position relative to short and long smoothed trends.
SMMA slopes	smma20_slope, smma120_slope	Measures whether the short and long trend baselines are rising/falling.
Trend	trend_5, trend_10	Captures multi-minute directional movement.
Recent range	distance_high_20, distance_low_20, close_position_20	Measures price location inside the recent 20-candle range.

6.1 Indicator Theory: SMMA

The project uses Smoothed Moving Average (SMMA) periods 20 and 120. SMMA gives more smoothing than a simple moving average and is used here to represent short-term versus long-term trend.

The implementation initializes the SMMA with the average of the first period, then recursively updates it with the previous SMMA and the new close. The resulting short/long relationship provides trend context to the classifier.

7. Target Engineering

For a candle at time t, the future return is determined from the close at t and the close ten minutes later.

future_return = (future_close - current_close) / current_close

The target classes are:

- BUY when future_return >= +0.10%
- SELL when future_return <= -0.10%
- HOLD otherwise

This target intentionally separates meaningful movement from small price changes. The 0.10% threshold is therefore a classification boundary, not a guarantee that an executed trade will earn 0.10%.

8. Machine Learning Theory

8.1 Why Classification

The system predicts a direction class rather than a continuous price. This makes it easier to evaluate whether the model selected the correct directional category and allows a probability distribution over classes.

8.2 Logistic Regression Baseline

Logistic regression provides a simple baseline for multiclass classification. It models a linear relationship between features and class log-odds. Its value in this project is diagnostic: if more flexible models cannot improve on a simple baseline, the additional complexity is not justified.

8.3 Random Forest

Random Forest combines many decision trees trained on bootstrapped samples and randomized feature subsets. The ensemble reduces the variance of a single tree and can capture nonlinear feature interactions.

8.4 Balanced/Frozen Random Forest Experiments

Alternative Random Forest variants were explored to test robustness to class distribution and model stability. These experiments helped compare different tree-based approaches before selecting the production configuration.

8.5 Histogram Gradient Boosting

The selected production model is Histogram Gradient Boosting. Boosting builds an additive sequence of weak learners, where each new learner focuses on the errors remaining from the previous ensemble. The histogram implementation bins continuous feature values to make tree split search efficient.

Learner 1 → residual errors
Learner 2 → focuses on remaining errors
Learner 3 → focuses on remaining errors
...
Final ensemble → combined nonlinear predictor

This model is suitable for the feature set because the relationships among returns, volatility, volume, SMMA variables and range-position features are not expected to be purely linear.

9. Production Model: HIST_GRADIENT_V2_1

Verified property	Value
Model version	HIST_GRADIENT_V2_1
Estimator	HistGradientBoostingClassifier
Feature count	26
Classes	BUY / HOLD / SELL
Artifact	data/models/production_model_wipro_v2_1.joblib
Validation	Joblib load verified successfully

The model artifact stores the trained estimator, the production feature list and model-version metadata. The live predictor reads these rather than hard-coding a second, potentially inconsistent feature list.

10. Training and Model Selection Procedure

1. Collect and prepare historical/live-derived candle data.
2. Compute the target using the future ten-minute close.
3. Generate the 26 engineered features.
4. Fit and compare candidate models.
5. Select the production configuration used for live prediction.
6. Persist the production model and feature metadata.
7. Load the same artifact in the live predictor.
8. Evaluate future predictions using actual subsequent market candles.

11. Prediction and Confidence Generation

At each new completed candle, the predictor builds a one-row feature DataFrame using the exact production feature list. The classifier returns a probability for each class.

```
probabilities = model.predict_proba(X)[0]

prediction = model.predict(X)[0]

confidence = probabilities.max()
```

The stored confidence is therefore the highest predicted class probability. It should be interpreted as model confidence, not as a guaranteed probability of profit. The reason for storing confidence is to support both filtering and later calibration analysis.

12. Prediction Persistence and Duplicate Prevention

Each prediction is stored with timestamp, symbol, model version, prediction class, confidence and entry price. The system checks whether the same timestamp and model version already exist before inserting a new record. A second in-process timestamp check prevents repeated processing of the same candle.

The final live validation showed zero duplicate predictions in the dashboard state reported during testing.

13. Outcome Evaluation Methodology

Evaluation deliberately happens after the prediction horizon. This prevents using future information at prediction time.

Prediction at time T
↓
Wait until T + 10 minutes
↓
Read actual future candle
↓
Calculate realized future return
↓
Convert actual return to BUY/HOLD/SELL
↓
Compare predicted class vs actual class
↓
Mark prediction evaluated

During the final live audit, the evaluator had processed most of the V2.1 queue. An intermediate check showed 200 total V2.1 predictions, 179 evaluated and 21 pending. The pending records correspond to recent predictions that did not yet have a complete ten-minute future window.

13.1 Why Confidence and Success Rate Must Be Analyzed Together

A strong model should not only produce different class probabilities; its confidence should also be informative. A practical calibration analysis therefore groups predictions into confidence bands and measures the realized success rate within each band.

Confidence band	Signals	Correct	Incorrect	Success rate
<50%	29	8	21	27.59%
50-60%	31	7	24	22.58%
60-70%	44	17	27	38.64%
70-80%	47	23	24	48.94%
80-90%	56	32	24	57.14%
90-100%	99	53	46	53.54%

This table should be populated from the evaluator before the final report is submitted. The present project evidence confirms confidence generation and evaluation, but the supplied console outputs do not contain the completed confidence-band statistics.

14. Full-Day and Next-Day Comparison

The review requirement specifically asks for one full day of recorded ML predictions followed by comparison with the next day. This is important because training accuracy or a static backtest number does not show how the live model behaves under a new session.

Metric	2026-08-31	2026-09-01
Predictions generated	17 evaluated V2.1 records in available sample	289 evaluated V2.1 records
Predictions evaluated	17	289
Correct predictions	3	137
Overall success rate	17.65%	47.40%
BUY success rate	N/A	0.00% (0/10)
HOLD success rate	22.22% (2/9)	55.41% (123/222)
SELL success rate	12.50% (1/8)	24.56% (14/57)
Average confidence	59.91%	78.05%
High-confidence success rate	To extract from evaluator	To extract from evaluator

The confidence-band results below were calculated from the evaluated V2.1 records. The model's confidence shows useful ranking behaviour in this sample, but it is not perfectly monotonic. This is why the paper-trading engine uses a confidence threshold as a filter instead of treating confidence as a guarantee of future profit.

Interpretation: the 2026-09-01 session showed materially stronger performance than the limited 2026-08-31 sample (47.40% vs 17.65% success). The average confidence also increased from 59.91% to 78.05%. However, the 2026-09-01 BUY class achieved 0/10 correct outcomes, so the current BUY-only paper-trading rule is not supported as a profitable directional rule by this sample. The result is presented as a validation finding, not hidden.

15. Paper Trading Implementation

Paper trading is a validation layer, not a live brokerage execution layer. The engine consumes V2.1 predictions, applies the confidence and signal rules, opens a virtual position if the rules are satisfied, and closes it at the exact ten-minute target candle.

Rule	Current value
Initial virtual capital	Rs 100,000
Quantity	100
Minimum confidence	60%
Allowed signal for entry	BUY
Horizon	10 minutes
Real broker orders	Disabled

15.1 Paper-Trading P&L Calculation

For a BUY position, the realized P&L is calculated as exit_price minus entry_price multiplied by quantity.

P&L = (exit_price - entry_price) × quantity

Observed validation trades included two historical SELL positions from an earlier configuration and one BUY position under the current BUY-only configuration. The final account state from the audit was Rs 99,959 cash and Rs -33 realized P&L.

Trade	Side	Entry	Exit	P&L
1	SELL	178.58	178.58	Rs 0.00
2	SELL	178.76	178.68	Rs +8.00
3	BUY	181.50	181.09	Rs -41.00
Total				Rs -33.00

16. Database and Live-Run Evidence

Audit item	Observed value at final audit snapshot
Ticks	501,098
Candles	24,278
V2.1 predictions	200+; exact live total continued increasing
V2.1 evaluated	179 at one audit snapshot
V2.1 pending	21 at the same snapshot
Paper trades	6 entries/exits
Paper positions	3 historical records, all CLOSED
Open positions	0
Latest audited V2.1 signal	2026-09-01 12:18:00 HOLD, 96.20%, Rs 181.36

17. Dashboard

The Streamlit application provides a user-facing view of the database and system state. It exposes Dashboard, Market, Predictions, Performance, Model, Paper Trading and System pages. The dashboard showed WIPRO market price, latest candle, AI signal, confidence, model version, candle counts, prediction counts and duplicate count.

The dashboard is intentionally not presented as the live-data integration itself. The backend terminals remain the primary evidence for Fyers connectivity and live processing.

18. Engineering Problems Encountered and Fixes

Problem	Diagnosis	Resolution
Duplicate Python processes	Parent/child process inspection showed two Python executables for one logical run.	Used process-tree inspection and taskkill/Stop-Process to clean up; added single-instance behavior to the predictor.
Invalid WebSocket token	Fyers returned code -300 at subscription.	Generated a fresh auth_code, exchanged it for access_token and verified token format.
Stale feed	Watchdog detected no target tick for about 30 seconds.	Added stale-feed detection and controlled socket recovery; distinguished inactive-

		market conditions from auth errors.
Candle gaps	Long missing minute intervals were observed.	Added one-minute continuity protection so the predictor skips discontinuous candles.
Duplicate predictions	Repeated polling could otherwise reprocess the same candle.	Timestamp/model-version duplicate check plus in-process last-timestamp check.
Dashboard dependency missing	Streamlit and matplotlib were not in requirements.txt.	Installed the dependencies and updated requirements.txt.
Submission compile issue	A diagnostic script contained invalid here-string syntax.	Removed the broken diagnostic file from the submission copy and re-ran compile validation.

19. Independent Technical Understanding

The review explicitly asks whether the candidate understands the implementation. The following points should be explained verbally during review or video:

- Why WebSocket is needed for real-time tick delivery.
- Why completed 1-minute candles are used rather than the currently forming candle.
- How ticks are aggregated into OHLCV candles.
- Why the target uses a ten-minute future return.
- Why the 0.10% threshold creates BUY/HOLD/SELL classes.
- Why Histogram Gradient Boosting was selected after model comparison.
- How predict_proba() produces class probabilities.
- Why maximum class probability is used as stored confidence.
- Why confidence is not the same as guaranteed trade success.
- Why predictions must be saved before the future outcome is known.
- Why the evaluator waits ten minutes.
- Why paper trading is used instead of live orders.
- Why data gaps should block prediction rather than be silently ignored.

20. Security and Submission Controls

The final submission copy was prepared separately from the live development environment. The final ZIP was checked for credentials and runtime files. No settings.py, .env, database, logs or compiled .pyc files were present in the submission copy at the final security check.

The actual access token, secret key and personal configuration must remain local and must never be placed in the public submission.

21. 15-Minute Video Demonstration Script

Time	What to show	What to explain
0:00–2:00	Project architecture	Objective, modules and why the dashboard is not the live-data source.
2:00–4:00	Fyers authentication and collector terminal	Auth flow, token exchange, WebSocket subscription.
4:00–6:00	Live tick stream	Actual WIPRO ticks arriving and being persisted.
6:00–8:00	Candle builder	How ticks become completed one-minute candles.
8:00–10:00	V2.1 predictor	26 features, model inference, BUY/HOLD/SELL and confidence.
10:00–12:00	Evaluator	Why predictions remain pending and how ten-minute outcomes mark them evaluated.
12:00–14:00	Paper trading	Entry criteria, exact-horizon exit, P&L and current position state.
14:00–15:00	Dashboard + results	Day-by-day success rates, limitations and final conclusion.

22. Final Results and Interpretation

The engineering pipeline has been demonstrated end-to-end using live Fyers data. The system is able to receive live market ticks, generate candles, compute the production feature set, produce V2.1 predictions with confidence, persist predictions, evaluate them later and simulate paper trades.

The final evidence now includes measured day-level results and confidence-band results. These should be reproduced in the 15-minute demonstration video.

23. Final Review Checklist

- Live Fyers WebSocket demonstrated during market hours.
- Ticks visible in backend terminal.
- Completed candles visible in candle-builder terminal.
- V2.1 model loaded and verified as HistGradientBoostingClassifier.
- 26-feature inference demonstrated.
- Prediction and confidence output demonstrated.
- Evaluator shown converting pending records to evaluated records.
- At least one full-day dataset and next-day dataset compared.
- Confidence-band success analysis included.
- Paper entry and exact-horizon exit demonstrated.
- Dashboard shown as a visualization layer.
- README and requirements updated.
- Final submission excludes credentials, database, logs and runtime environment.
- 15-minute video recorded.

24. Limitations

- The current live validation history is limited compared with a production trading deployment.
- The paper-trading sample is small and includes a negative realized P&L.
- Confidence is model-derived probability and is not a guarantee of trade profitability.
- Real execution costs, slippage and market impact are not identical to paper execution.
- The one-symbol production demonstration should not be generalized to all stocks without additional validation.

25. Conclusion

The SSG AI/ML Trading System provides a practical implementation of the requested workflow: live Fyers integration, background processing, supervised machine-learning prediction, confidence generation, future-outcome evaluation, paper trading and dashboard monitoring. The implementation also includes practical safeguards for duplicate predictions, incomplete candles, data gaps and credential separation.

For final submission, the only critical evidence still requiring insertion is the exact evaluator-generated full-day and next-day success-rate comparison. That metric must be computed from the actual stored predictions and outcomes and should be shown both in the report and in the 15-minute video.

Appendix A — Production Runtime Commands

python -m auth.fyers_auth

python -m auth.exchange_token

python -m fyers.fyers_collector

python -m engine.candle_builder

python -m ml.live_predictor_v2_shadow

python -m ml.live_evaluator

python -m paper_trading.engine

python -m streamlit run .\dashboard\app.py

Appendix B — Final Verified Production Model

data/models/production_model_wipro_v2_1.joblib

Verified model version: HIST_GRADIENT_V2_1

Verified estimator: HistGradientBoostingClassifier

Verified feature count: 26